2025-2026

MARMARA UNIVERSITY

Department of Computer Science & Engineering

CSE 1242 - Introduction to Programming II

Term Project

MAY 17, 2026

23: 59

1.Project Introduction

The **Ghost Hunter Inc.** project is a technical demonstration of object-oriented programming (OOP), event-driven systems, and real-time rendering using the JavaFX framework. The project aims to provide an engaging 2D environment where students can apply theoretical concepts of event-driven applications, collision detection, and scene graph management.

The narrative serves as the functional wrapper for the game's mechanics. Players assume the role of a "Ghost Containment Specialist" for a high-tech startup, **Ghost Hunter Inc.**, tasked with clearing paranormal infestations from localized haunted regions.

The core thematic hook is that the map appears empty without any paranormal entity to the naked eye. The technical challenge for the player is to utilize an active scanning mechanic to reveal and vacuum entities that are logically present in the data but visually absent from the render in a limited time.

The game must consist of (at least) 3 levels.Each level has different game environment, paranormal entities and time limitation. Each entity that is captured will provide different scores. Also, (at least) in each level, 3 types of tokens that provide different features to the player will be available to obtain in the game. Following content provide detailed expalanation for each component and mechanics.

2. Game World and Environment Design

The game consist of 3 different levels where each level has different map design with different theme, time limit, paranormal entity types and counts. <u>The player must vacuum all the paranormal entities in the map without exceeding limited time determined for that level in order to finish the current level and be able to get to the next level.</u>

Also, each level must have a playable area to constraint both paranormal entities and player to move in the map. Thus, each map must be designed to allow movement only in this playable area for paranormal entities and the player. Also, token spawn locations must be in the playable area either.

Lastly, for each level, Hunter's (playable character) health bar that represents it's health and vacuum bar that represents it's ability to scan/vacuum the entities must be visible on the screen left and right side visibly. In the middle of both bars, there should be a text representing the score and remaining time for the current level.

Health Bar (Right Side)

- **Shape:** A tall, vertical rectangular container with a thick black border on the right side of the screen.
- **Color:** The fill is a **vibrant, solid red**, traditionally signifying vitality.

- **Label:** The word "HEALTH:" is placed directly above the bar in a white, stylized, "spooky" or hand-drawn font that matches the game's theme.

Vacuum Bar (Left Side)

- **Shape:** Mirrored to the health bar, it is a tall, vertical rectangle with a thick black border on the left side of the screen.
- **Color:** The fill is a **bold purple**, which increase/decrease with the "vacuum" mechanic.
- **Label:** The word "VACUUM:" sits above the bar in the same white, stylized thematic font.

Score and Timer (Top Center)

- **Text Style:** Both elements use a the same clean, spooky" or hand-drawn font white font.
- **Score:** Displayed as "SCORE: 0". It is positioned at the top to give the player immediate feedback on their progress.
- **Timer:** Located directly beneath the score (e.g., "1:27"). The use of a simple colon format (Minutes:Seconds)

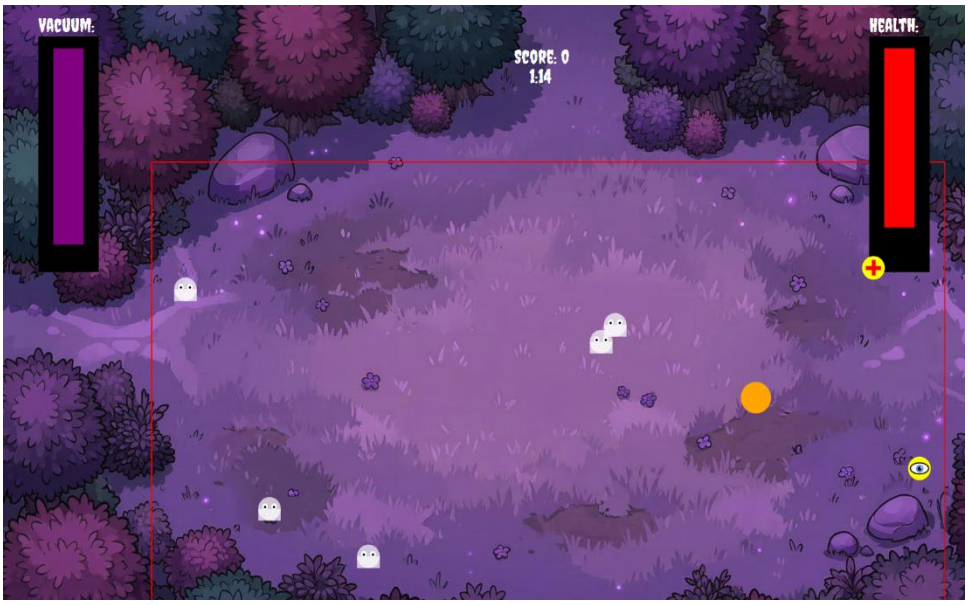
Each level scales in difficulty and complexity through the following variables:

- **Environmental Design:** Unique maps and aesthetic themes, which you can design by your own choice.
- **Entity Variations:** Specific types and quantities of ghosts/entities.
- **Time Constraints:** Decreasing time limits to heighten the challenge.

To ensure flexibility and ease of modification, all level-specific properties—including map layouts, entity counts, and time limits—are dynamically loaded from an external **.txt file**.

2.1. Level 1:

First level has basic and easy features in order to get user familliar to the environmet and game mechanics. Thus, difficulty of the gameplay is also considered easy by setting the paranormal entity type only as Ghost and spawn only 5 ghosts for this level.

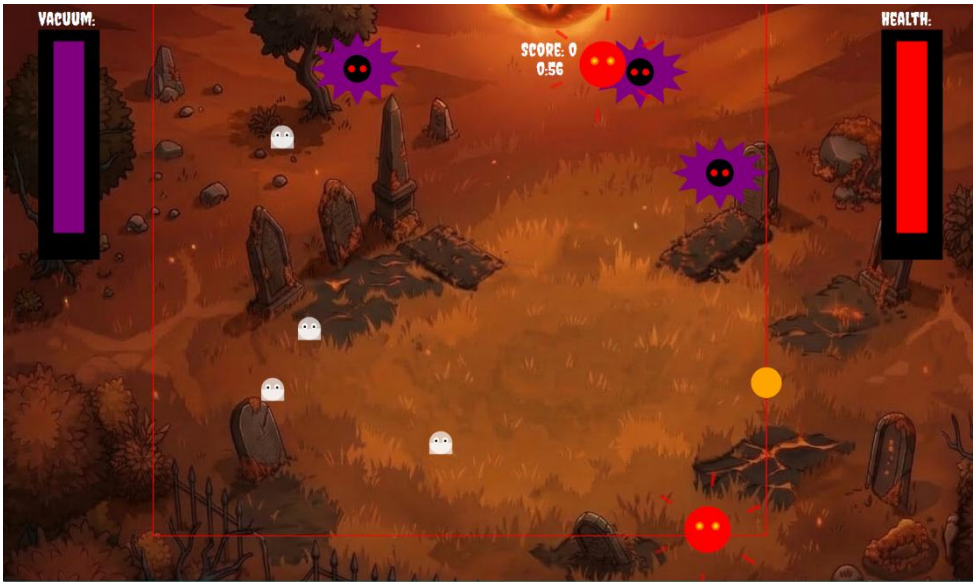


- Environment Theme: Deep purples, magentas, and soft indigo. (You can design your own background environment)
- Time Limit: 60 seconds
- Entity types and Counts:
    - Ghosts: 5
    - Ripper: 0
    - Wisp: 0

2.2. Level 2:

- Environment Theme: Cold teals, misty blues, and deep shadows. (You can design your own background environment)
- Time Limit: 120 seconds
- Entity types and Counts:
    - Ghosts: 4
    - Ripper: 3
    - Wisp: 0

2.3. Level 3:



- Environment Theme: Aggressive oranges, burnt reds, and charcoal blacks. (You can design your own background environment)
- Time Limit: 180 seconds
- Entity types and Counts:
    - Ghosts: 4

- Ripper: 3
- Wisp: 2

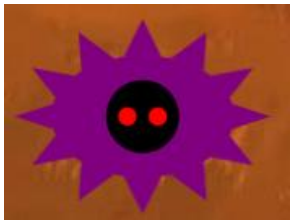
Note that a certain key must be used as a "cheat key" to display the playable area boundries and entities in the map.

3. Character and Entities

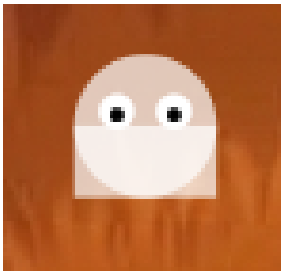
An example of character and entity design is given. However, you can implement your own design for character and entities. Note that they must be implemented using geometric shape libraries in the Java FX framework.

- **3.1. The Hunter (Player):**
  - **Visual Components:** Body (Circle) and a Semi- transparent red triangle with a given dynamic range.

- **3.2. Paranormal Entities (Enemies):**
  - **Ripper:** Spiky purple-bodied, random, and menacing.



  - **Ghost:** Classic white ghost, random speed, semi-transparent.



- **Wisp:** Random speed, big, consisting of glowing concentric circles with revolving rectangles around his body.

4.Game Mechanics

4.1. Hunter Movement:

- Hunter must move different directions seamlessly (without any stuttering) with given keyboard for that directions.

  W: Up

  A: Left

  D: Right

  S: Down

- If hunter reaches one of the borders of the playable area, it can't further proceed in that border's direction.
- Whether moving or standing in the game, if hunter overlaps with a paranormal entity, hunter's geometrical shape color must turn red and its health must drop in the health bar.

4.2. Hunting and Detection Mechanism:

- By using SPACE key in the keyboard, scanner can be activated and stays active as long as SPACE key is pressed.

- Activating scanner makes the scanner triangle visible as long as SPACE key is pressed
- As user activates and uses the scanner (presses SPACE), the scanner bar on the left side of the corner (that states the energy of the scanner) will drop until it reaches 0.When it reaches 0, scanner will deactivate itself and scanner triangle will be invisible again.
- While user is not using the scanner(not pressing SPACE), scanner bar will be consistently increasing up until its full potential. As long as scanner bar is filled with energy (no matter it's amount), it can be activated.
- If a paranormal entity enters the range of scanner triangle, it becomes visible.However, it's shapes' original colors will change to a combination of white/gray as long as it's in the scanner range.
- Entities inside the scanner (or vacuum) triangle will shrink consistently as long as they stay inside the triangle. (Vacuum effect)
- Vacuumed entities increase hunter's score. Each entitiy type has different score
    - Ghost: 10 points
    - Ripper: 20 points
    - Wisp: 30 points

4.3. Paranormal Entity Movement

- Each paranormal entity will spawn random locations in the playable area when the level starts.
- Each entity starts with a random direction to move.
- If an entity reaches one of the borders of the playable area, it must bounce back and proceed with the symmetrical path of its incoming path.
- Maximum health of a hunter is considered as 100.
- <u>If one of the entitites overlaps with the hunter, hunter's health must be consistenly decreasing 2 health units throughout the overlap</u>. Note that since paranormal entities' speeds are random, if a slow entity collides with hunter while hunter is not moving, it's going to give more damage since collision duration will be more than the one with a fast entity.

4.4. Token Design and Mechanics

Three types of token exist in the game that give different benefits to the player for that level. These are, Health Token, Range Token and Eye Token.

- <u>Each token must be designed using Java FX geometric classes.</u>
- <u>You can customize your own token types and designs other than the given tokens in this document.</u>

- Tokens will be generated in the map each 5 seconds.Token locations and token types will be selected randomly for each generation.
- Only two tokens are allowed to exist simultaneously. If number of current tokens in the map is two, new tokens won't be generated.

4.4.1. Heath Token

- Health token has the ability of increasing player's health.
- After health token is taken by the hunter, its current health will increase and simultaneously, its health bar will display this as an animated increase too.
- The amount of restorated health that health token provides will be taken from the text file. You can determine the amount of increase that the token provides.

4.4.2. Range Token

- Range token has the ability of increasing the hunter's scanner/vacuum range.
- After it is taken by the hunter, the scanner's triangle must be wider by 20 pixels from both equal edges.
- The amount of the increase in the scanner's triangle will be taken from the text file. You can determine the amount of increase that the token provides.

4.4.3 Eye Token

- Eye token has the ability of making every paranormal entitiy visible in the map for a certain amount of time.
- After it is taken by the hunter, each paranormal entity will be visible for some seconds. After the given time is over, they will become invisible again.
- The amount of time that entities will be visible after the hunter acquires the eye token will be taken from the text file. You can determine the visibility time by yourself, however, 5 seconds is suggested.

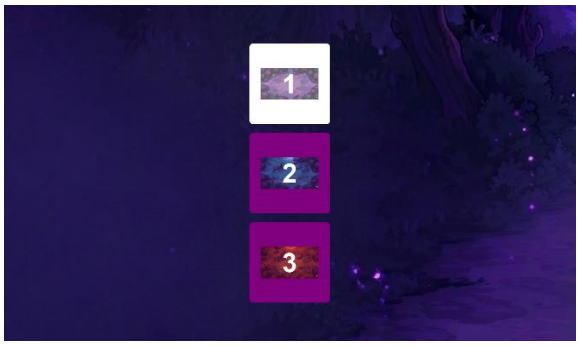
5. User Interface (UI) and Menu System

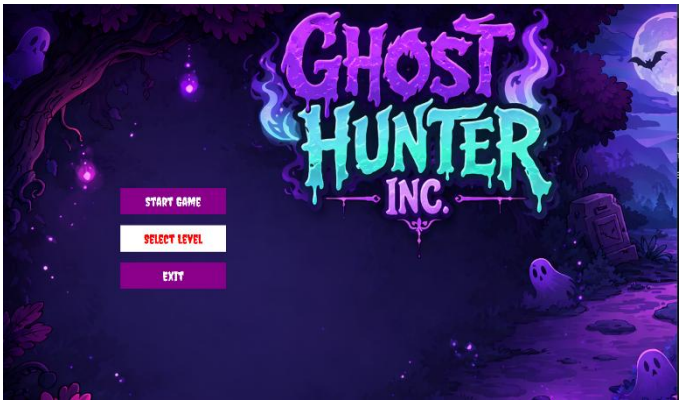
5.1. Main Menu:

- Using a background image with 3 buttons. Each buttons consist of a rectangle background and a white text. While buttons are in the clicked state, background color must turn white and text color must turn red

- **Start Game Button:** Starts the game from the first level
- **Select Level Button:** Selects a level to start the game with
- **Exit Game Button:** Terminates the program and exits the game

If player selects the “Select Level” button, a page that contains small version of each levels’ map will be displayed as images. Each map images has a purple background rectangle that turns red when user is in the clicking state to that map. When player’s mouse is hovering-over the maps, purple tone of the background of the button that mouse is currently on must be a little more lighter. If the user clicks a button, its background must become white and when it’s released, game must start with that map.

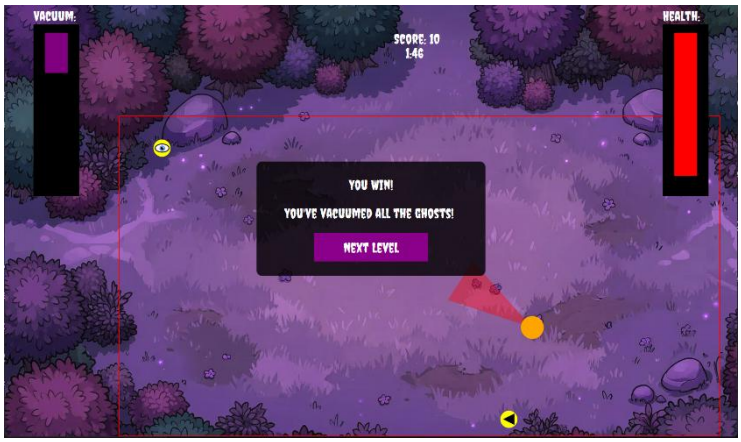




5.2. Scene Switching

- After the end of each level, game must be stopped (timer, paranormal entities and user movement etc.). According to the reason of the ending, a panel must display some options.

5.2.1. Win Panel:

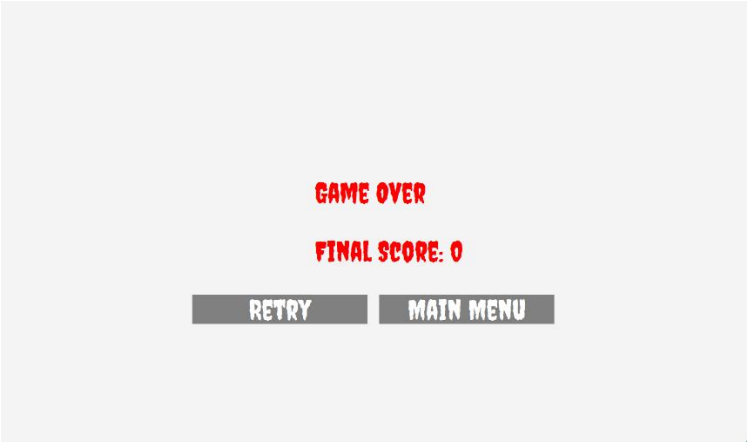


If the user passes the current level by successfully vacuuming all the entities, panel will display a message such as “You Win!" with a subtext with “You've vacuumed all the ghosts!” .

Panel will have semi-transparent black backgrund. Lastly, it has a button to go to the next level when pressed and released. When button is in the clicked state, its background will turn red.

5.2.2. Lose Panel:

If the user health is diminished to 0 or the time is up before user is vacuumed , the game will stop and a white background that has a “Game Over” message will appear. In this panel, player’s final score and buttons for retrying the current level and exiting to the main menu is included. Color changing on the clicked state and action when released is also valid here.



6. Text File

As stated above, your game must implement certain configurations for some features. These configurations will be read from a text file called “config.txt”. An example for config.txt file is given below:

Note: You can start by hardcoding each configurations and after you make sure that it works, you can convert it from hardcoded to text file reading version.

```
maximum_health: 100
maximum_vacuum: 100
entity_damage: 10
vacuum_decrease: 2
vacuum_increase: 5

level_1_playable_area_x: 200
level_1_playable_area_y: 200
level_1_playable_area_width: 1050
level_1_playable_area_height: 560
level_1_time: 60
level_1_ghosts: 5
level_1_rippers: 0
level_1_wisps: 0

level_2_playable_area_x: 200
level_2_playable_area_y: 100
level_2_playable_area_width: 1000
level_2_playable_area_height: 560
level_2_time: 120
level_2_ghosts: 4
level_2_rippers: 3
level_2_wisps: 0

level_3_playable_area_x: 200
level_3_playable_area_y: 0
level_3_playable_area_width: 800
level_3_playable_area_height: 700
level_3_time: 180
level_3_ghosts: 4
level_3_rippers: 3
level_3_wisps: 2

health_token_increase: 20
vacuum_token_increase: 20
eye_token_duration: 5
```

6.1. Global Player Stats

These variables define the base attributes of the player character across all levels.

- **maximum_health:** The total HP capacity of the player.
- **maximum_vacuum:** The maximum energy or fuel capacity for the vacuum tool.
- **entity_damage:** How much health the player loses when colliding with or being attacked by an enemy.
- **vacuum_decrease:** The rate at which the vacuum's energy is consumed during use.
- **vacuum_increase:** The rate at which the vacuum recharges (likely when idle or using a specific mechanic).

6.2. Level-Specific Parameters

These settings repeat for Level 1, 2, and 3, allowing you to increase the difficulty as the player progresses.

6.2.1. Playable Area (The Boundary)

- **playable_area_x / y:** The starting coordinates (top-left corner) of the active game zone on the screen.
- **playable_area_width / height:** The total dimensions of the game world. Notice that Level 3 is narrower but taller than Level 1.

6.2.2. Level Objectives & Difficulty

- **time:** The countdown limit (in seconds) the player has to complete the level.
- **ghosts:** The number of standard ghost enemies to spawn.
- **rippers:** The number of "Ripper" type enemies (starts at 0 in Level 1, increasing to 3 by Level 3).
- **wisps:** The number of "Wisp" type enemies (only appears in Level 3).

6.3. Items & Power-Ups

These define the effectiveness of the pick-up items found within the levels.

- **health_token_increase:** How much HP is restored when picking up a health item.
- **vacuum_token_increase:** How much vacuum energy is restored when picking up a vacuum refill.
- **eye_token_duration:** How long the "Eye" power-up effect lasts (likely a vision-enhancing or reveal mechanic).

7.Submission Instructions

7.1. Zip File: Submit a ZIP file named Student1ID_Student2ID_Student3ID_Project.zip containing:

- Source code with detailed comments.
- A 5-10 page project report.

7.2. Deadline: 17.05.2026, 23:59

7.3.Demo Sessions: Demonstrate gameplay, hunter movement, paranormal entity design and movement, vacuum and scan mechanics, scale-up/down mechanics, color change and animations of vacuumed entities, health bar and vacuum bar mechanics etc.

7.4.Project Report Requirements

- Cover Page: Title, group member details, course information.
- Problem Definition: Describe the game in 1-2 paragraphs.
- Implementation Details: UML diagrams for core classes.
- Test Cases: Screenshots of gameplay with explanations.

7.5.Notes

- You cannot use Scene Builder for the project.
- Code Comments: Include your name, ID, and purpose of each class/method.
- Plagiarism: FF grade for copied work.
- Late Submissions: Not accepted.
- You will be given a video on how to play the game.